

Database Management system

Types of databases

1. SQL(RDBMS)
 - a. Structured data
 - b. data should be secure.
 - c. Transaction control is needed.
example : oracle, mysql,sql server, postgrSQL
2. NoSQL
 - a. unstructured
 - b. data may not be secure.
 - c. No transaction control
examples: mongodb, CouchbaseDB, Cassandra
3. GraphDB
 - a. unstructured
 - b. display should be in graph format
examples : NEO4j
4. Memory databases
 - a. if the size of data is small, and faster access is needed.
example MemDB, VoltDB
5. Disk based databases.
 - a. These databases are available on the same machine on which it is installed
examples : SQLite3, MS-ACCESS

uses of SQL databases

1. Sharing of data is easy
2. data is secure, it supports transaction control
3. managing data becomes easy.
4. retrieval of data is easy.

Types of statements available

Types of statement		
DQL	Data Query Language	select
DCL	Data Control Language	grant , revoke
TCL	Transaction control language	commit, rollback, savepoint
DDL	Data definition language	create, alter,truncate,drop
DML	Data Manipulation language	Insert, update delete

ACID property

1. Atomicity---If a transaction contains many steps, then either all will happen or none will happen, entire transaction will get executed as a single unit
2. consistency--→ after every transaction the data will be correct state
3. isolation--→ the intermediate changes in the transaction are not visible to all users, is called as isolation
4. durability--→ the correctness data after every transaction, will be there for longer duration.

Data gets stored in table format in SQL

1. Table is also called as relation.
2. Columns → attribute/ fields/ columns
3. record → record/ row/ tuple

Install mysql

<https://dev.mysql.com/downloads/installer/>

To check whether server is running(Windows)

1. go to start button > services
2. find MySQL80, and check the status(running)

To open client

3. go to start button> mysql command line client, Mysql workbench
or
 1. open cmd prompt
 2. mysql -u root -p
password: <while typing password it will not be visible>

To uninstall mysql

1. go to start button > type **add or remove program**
2. uninstall every application, with name starts with mysql
3. delete these following folders also
 - a. C:\Program Files\MySQL – Contains the core application files
 - b. C:\Program Files (x86)\MySQL
 - c. C:\ProgramData\MySQL (In file explorer> view >show view>hidden elements)
 - d. C:\Users\<YourUsername>\AppData\Roaming\MySQL

MySQL store data internally in the form files, called as table space,
It stores data in various types of files

4. control file → these files are used to store metadata
5. data file → these files are used to store data
6. redolog files → these files are used to store all changes happening to data, and can be used at the time of commit, and rollback

To create database in mysql

Create database iacsdacaug26

create database if not exists <name of the database>

create database if not exists dbdai0925

To load data from SQL file

mysql> source <filename>

To see the list of databases

Show databases;

To see the list of tables

Show tables;

To change the database

Use iacsdacaug26

Operators in mysql

Arithmetic operators	+, -, /, *, %
logical operators	and or not
relational operators	=, !=, >=, <=, >, <

Operators

[not] between ... and ...	To check the range of values we use between and operator, the given values are inclusive
[not] in	to check multiple values = with either or, the use in
is [not] null	to check null values we use this operator
[not] like	to check the pattern in the field we use like and not like operator in this % symbol represents 0 or more characters and _ represents exactly one character

DQL --→ Data query language

1. to display all fields and all rows from emp table
select * from emp
2. to display empno,ename for all employees
select empno,ename from emp;
3. to display all employees with sal > 2000
select * from emp
where sal>2000;
4. to display all employees with job=salesman
select * from emp
where job='Salesman'
5. to display all employees joined on '1981-12-03'
select * from emp
-> where hiredate='1981-12-03'

Note: to improve the efficiency of the query
condition which is true less number of times should be before, in and operator
Condition which is false less number of times should be written first, in or
operator

6. to display all employees with job=salesman
and deptno=10
select * from emp
where job='Salesman' and deptno=10;

7. display all employees working as CLERK and sal >1500

select * from emp
where job='CLERK' and sal>1500

8. display all employees with sal=1500 or 2000 or 3000

select * from emp where sal=1500 or sal=2000 or sal=3000;	select * from emp where sal in (1500,2000,300)
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9. display all employees with sal not either 1500 or 2000 or 3000
select * from emp
where sal not in (1500,2000,300)

10. display all employees with job clerk, salesman or manager

select * from emp
where job in ('clerk','salesman','manager')

11. display all employees who work in deptno 10 or 20 and sal > 1500

select * from emp
where deptno in (10,20) and sal > 1500;

12. display all employees with sal >=1500 and <= 2500

select * from emp
where sal between 1500 and 2500;

13. display all employees with sal <=1500 and >= 2500
1500-----2500

select * from emp
where sal not between 1501 and 2499;

14. display all employee with commission is null
select * from emp
-> where comm is null;

15. display all employee with commission is null

select * from emp
 -> where comm is null;

15. to display all names start with either a or m and ends with n or r

select * from emp where ename like 'a%n' or ename like 'a%r' or ename like 'm%n' or ename like 'm%r';	select * from emp where ename regexp '^[am].*[nr]\$';
---	--

15. to find all names starts with A

select * from emp where ename like 'A%'	select * from emp where ename REGEXP '^A'
--	--

16.

17. to find all names ends with N

select * from emp where ename like '%N'	select * from emp where ename regexp 'N\$'
--	---

18. to find all names which has E at 2nd last position

select * from emp where ename like '%E_'	select * from emp where ename regexp 'E.\$'
---	--

19. to display all employees with name starts with either A or M

select * from emp where ename like 'A%' or ename like 'M%'	select * from emp where ename regexp '^[AM]'
--	---

20. to display all employees with name starts with A and ends with N
 or starts with m and ends with R

select * from emp where ename like 'A%N' or ename like 'M%R'	select * from emp where ename regexp '^A.*N\$ ^M.*R\$'
--	---

21. display all employees with name starts with A and l is 3rd position and ends with
 N or starts with J and ends with either S or N

select * from emp where ename like 'A_L%N' or ename like 'j%S' or ename like 'j%N'	select * from emp where ename regexp '^A.L.*N\$ ^J.*[SN]\$'
---	---

22. display all employees with name N occurs at either 2nd position or 3rd position or starts with J and N at 3rd last position

select * from emp where ename like '_N%' or ename like '__N%' or ename like 'J%N__';	select * from emp where ename regexp '^..?N ^j.*N..\$'
---	---

23. display all employees with name does not start with either J or M

^[^jm]

select * from emp where ename not like 'J%' and ename not like 'M%'	select * from emp where ename regexp '^[^JM]'
---	--

- 24 display all employees with r or l exists at position between 3 to 6th position

select * from emp where ename regexp '^.{2,5}[rl]'

24. display all employees empno,name,sal,comm,and netsal, to calculate
netsal=sal+comm

select empno,ename,sal,comm,sal+ifnull(comm,0) from emp;

25. display all employees with name has l at 3rd position and n at last position but
between l and n minimum 1 character has to be there

select * from emp where ename regexp '^..l.[nr]\$';

Regular expression

^	it will check the pattern at the beginning
\$	it will check the pattern at the end
[a-zA-Z]	will match with any alphabet
[0-9]	will match with any one digit
.	match with any one character
[^0-9]	any character other than 0-9
*	0 or more occurrences
+	1 or more occurrences
?	0 or 1 occurrences
{m}	exactly m occurrences
{m,n}	minimum m occurrences or maximum n occurrences
abc pqr mns	to match with either abc or pqr or mns

ba*n	bn,ban, baaaaaaaaaan,this is baaaaaaaa
^[0-9]{5}	12345, 123456789

[0-9]\$	this is 123, this is help
^or	origami, organ, normal
or\$	taylor, minor
A.?b	Ab , Axb

1. to display empno,name,sal for all employees
select empno,name,sal from emp;
2. to display empno,name,sal,comm and netsal for all employees
where netsal = sal+comm
select empno,ename name,sal salary,comm,sal+ifnull(comm,0) "net sal" from
emp;